

TrES-5b

R-2378

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_140714 · created 2026-08-20T00:58:03+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456852.8669 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.054
Transit depth (Rp/Rs)^2	3.4 +/- 1.99 [%]
Orbital Inclination (inc)	89.63 +/- 2.88
Airmass coefficient 1 (a1)	0.606 +/- 0.068
Airmass coefficient 2 (a2)	0.4 +/- 0.53
Scatter in the residuals of the lightcurve fit is	11.25 %
Best Comparison Star	#2 - [357, 448]
Optimal Aperture	1.82
Optimal Annulus	10.8
Transit Duration (day)	0.0873 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.054 vs 0.142 (deviation 0.78σ)
Duration fit vs published	0.0873 d vs 0.0758 d (deviation 1.53σ)
Mid-time O–C	-4.2 min
Depth SNR	3.4
Residual scatter	11.25 %

Light curve

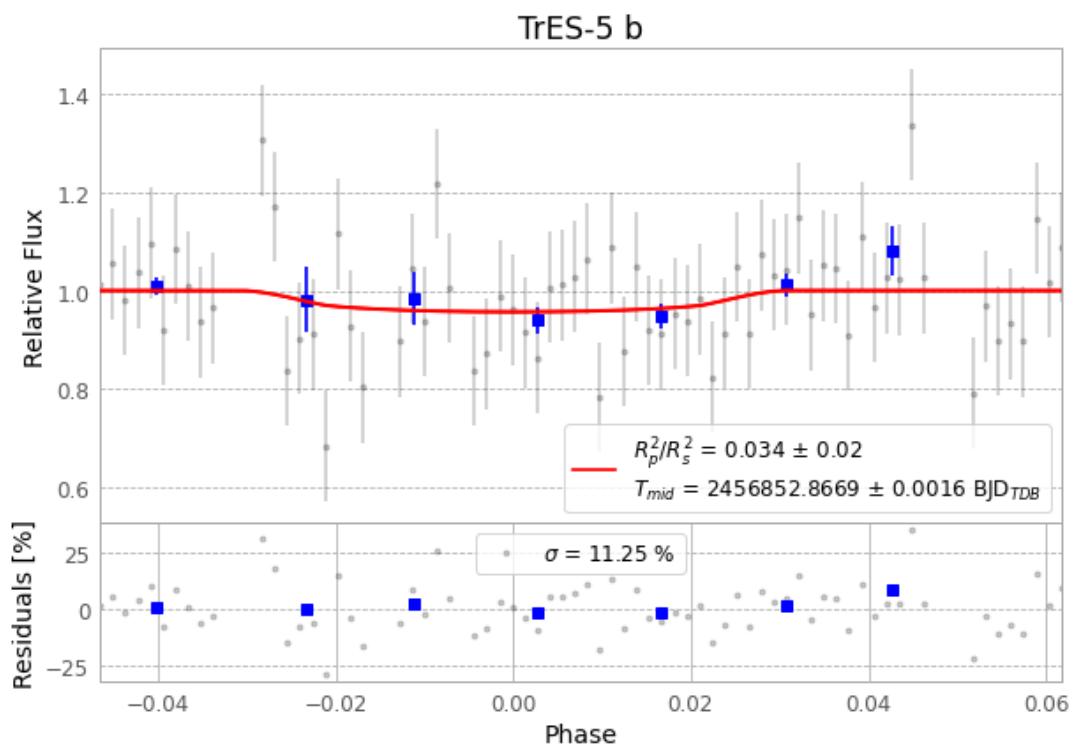


Figure 1 — FinalLightCurve_TrES-5 b_2014-07-14.png (SHA-256 b2b2f8b40d968beb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 312], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0270181eeb79fb76...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

78 FITS frames (51 MB), TRES-5140714070612.FITS → TRES-5140714105712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-140714.log (292d9210ba08...), FinalLightCurve_TrES-5 b_2014-07-14.png (b2b2f8b40d96...), FinalLightCurve_TrES-5 b_2014-07-14.csv (f718a4d778e0...)

Compute resources

Wall time 143.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:58Z.